RC RECTANGULAR COLUMN CHECK (EC2)

1. DESIGN BASIS

EN 1992-1-1. Fibres, strain pivots A/B/C; neutral axis rotated to the MEd direction (P–Mθ).

Flexure: ray from 0 through (MEd, NEd) to the P–Mθ curve (UR_M). fcd and shear (links):

EN 1992-1-1:2023 (5.3), §8.2.3.

x along b, y along h; N positive in compression. Annex A.n is calculated at its link ▸.

2. INPUTS

2.1 Materials

Characteristic concrete cylinder strength $f_{ck} := 35 \cdot \mathrm{MPa}$

Partial factor for concrete $\gamma_C := 1.5$

Sustained-load factor (5.3): 0.85, or 1.0 if loaded after 90 days $k_{tc} := 0.85$

Yield strength of longitudinal bars $f_{yk} := 500 \cdot \mathrm{MPa}$

Yield strength of links $f_{ywk} := 500 \cdot \mathrm{MPa}$

Ductility class of the links: 1 = class A, 0 = class B or C $\delta_A := 0$

Partial factor for reinforcement $\gamma_S := 1.15$

Elastic modulus of reinforcement $E_s := 200 \cdot \mathrm{GPa}$

Ultimate concrete strain (Table 3.1) $\varepsilon_{cu2} := 0.0035$

Concrete strain at peak stress $\varepsilon_{c2} := 0.002$

Design limit of steel strain, 0.9·εuk $\varepsilon_{ud} := 0.045$

2.2 Section and reinforcement

Section width (along x) $b := 500 \cdot \mathrm{mm}$

Section depth (along y) $h := 800 \cdot \text{mm}$

Nominal cover to the links $c_{nom} := 40 \cdot \text{mm}$

Longitudinal bar diameter $\phi := 25 \cdot \text{mm}$

Bars along each b face, corners included (≥ 2) $n_b := 4$

Bars along each h face, corners included (≥ 2) $n_h := 6$

Tie (link) diameter $\phi_w := 10 \cdot \text{mm}$

Tie spacing along the column $s := 150 \cdot \text{mm}$

Tie legs are arranged automatically (EC2 9.5.3(6): bars ≤ 150 mm from a restrained bar).

2.3 Integration

Concrete fibres along each side (nf × nf grid) $n_f := 30$

2.4 Design actions (ULS)

Axial force, compression positive $N_{Ed} := 3000 \cdot \text{kN}$

Moment about x (compression on +y face when positive) $M_{Ed.x} := 600 \cdot \text{kN} \cdot \text{m}$

Moment about y (compression on +x face when positive) $M_{Ed.y} := 300 \cdot \text{kN} \cdot \text{m}$

Shear force along y $V_{Ed.y} := 400 \cdot \text{kN}$

Shear force along x $V_{Ed.x} := 200 \cdot \text{kN}$

3. DESIGN VALUES

Effective strength factor (5.4), fck,ref = 40 MPa $$\eta_{cc} := \min\left(\left(\frac{40 \cdot \text{MPa}}{f_{ck}}\right)^{\frac{1}{3}}, 1\right) = 1$$

Design compressive strength (5.3) $$f_{cd} := \frac{\eta_{cc} k_{tc} f_{ck}}{\gamma_C} = 19.83 \cdot \text{MPa}$$

Design yield strength of longitudinal bars $f_{yd} := \frac{f_{yk}}{\gamma_S} = 434.8 \cdot \text{MPa}$

Design yield strength of links $f_{ywd} := \frac{f_{ywk}}{\gamma_S} = 434.8 \cdot \text{MPa}$

Design yield strain of longitudinal bars $\varepsilon_{yd} := \frac{f_{yd}}{E_s} = 0.00217391$

4. SECTION GEOMETRY

4.1 Bars

▸ **A.1** Bar layout and coordinates · see page 11

Bar spacing (centres) along b $s_x = 125 \cdot \text{mm}$

Bar spacing (centres) along h $s_y = 135 \cdot \text{mm}$

Clear spacing along b, ≥ max(φ, 20 mm) (8.2)

$s_x - \phi = 100 \cdot \text{mm} \geq \max(\phi, 20 \cdot \text{mm}) = 25 \cdot \text{mm}$ ✓ PASS

Clear spacing along h

$s_y - \phi = 110 \cdot \text{mm} \geq \max(\phi, 20 \cdot \text{mm}) = 25 \cdot \text{mm}$ ✓ PASS

Total number of bars $n_{bar} = 16$

4.2 Ties (EC2 9.5.3)

Tie diameter ≥ max(6 mm, φ/4), 9.5.3(1)

$$\phi_w = 10 \cdot \text{mm} \geq \max\left(6 \cdot \text{mm}, \frac{\phi}{4}\right) = 6 \cdot \text{mm}$$ ✓ PASS

Max. tie spacing min(20φ, b, h, 400 mm), 9.5.3(3)

$s = 150 \cdot \text{mm} \leq \min(20\phi, b, h, 400 \cdot \text{mm}) = 400 \cdot \text{mm}$ ✓ PASS

Bar spacing on a b face, 9.5.3(6) $s_x = 125 \cdot \text{mm} \leq 150 \cdot \text{mm}$ ✓ PASS

Bar spacing on an h face $s_y = 135 \cdot \text{mm} \leq 150 \cdot \text{mm}$ ✓ PASS

▸ **A.2** Tie leg arrangement (EC2 9.5.3(6)) · see page 12

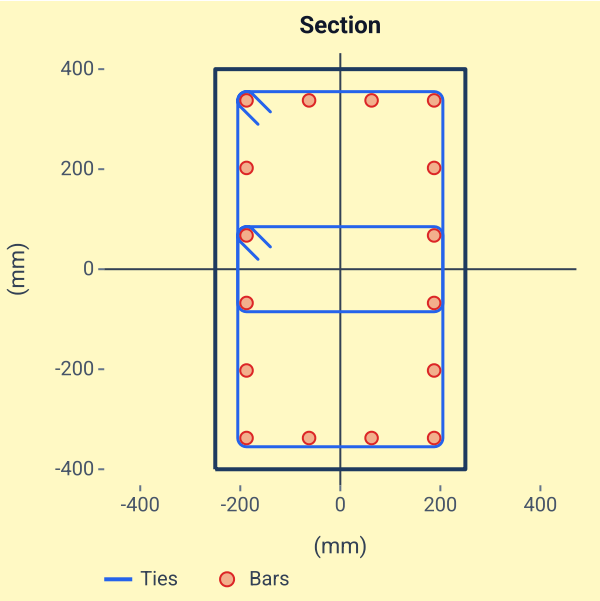
Full-depth tie legs parallel to y (resist VEd,y) $n_{w.y} = 2$

Full-width tie legs parallel to x (resist VEd,x) $n_{w.x} = 4$

4.3 Concrete fibres and section

▸ **A.3** Concrete fibres · see page 13

▸ **A.4** Tie drawing (data of the section figure) · see page 14



4.4 Steel areas

Area of one bar $A_b := 0.25\pi\phi^2 = 491 \cdot \text{mm}^2$

Gross concrete area $A_g := bh = 400000 \cdot \text{mm}^2$

Total longitudinal steel area $A_s := n_{bar} A_b = 7854 \cdot \text{mm}^2$

Minimum steel, EC2 (9.12N)

$$A_s = 7854 \cdot \text{mm}^2 \geq \max\left(\frac{0.1 N_{\text{Ed}}}{f_{\text{yd}}}, 0.002 A_g\right) = 800 \cdot \text{mm}^2$$ PASS

Maximum steel, EC2 9.5.2(3) $$A_s = 7854 \cdot \text{mm}^2 \leq 0.04 A_g = 16000 \cdot \text{mm}^2$$ PASS

5. FLEXURAL CHECK

5.1 Biaxial resistance: P–Mθ at the moment direction θ

The neutral axis is turned (angle β of its normal) until the resisting moment points along θ;

this gives the P–Mθ curve at θ. No reduction to the principal axes.

▸ **A.5** Strain plane, stress laws and section resultants · see page 17

Resultant design moment $$M_{Ed} := \sqrt{M_{Ed.x}^2 + M_{Ed.y}^2} = 670.8 \cdot \text{kN} \cdot \text{m}$$

Direction of the compression face (MEd,y → +x, MEd,x → +y) $$\theta_M := \text{angle}\left(M_{Ed.y}, M_{Ed.x}\right) = 63.4^\circ$$

▸ **A.6** P–Mθ diagram data (neutral-axis sweep) · see page 19

Curve scaled by its maxima (the ratio does not depend on the scale)

$$n_v := \frac{N_{\theta v}}{\max(N_{\theta v})}$$

$$m_v := \frac{M_{\theta v}}{\max(M_{\theta v})}$$

Utilisation (as MBColumn): ray from the origin through (MEd, NEd) to the P–Mθ curve.

Design point scaled the same way $$n_{Ed} := \frac{N_{Ed}}{\max(N_{\theta v})} = 0.2748$$

$$m_{Ed} := \frac{M_{Ed}}{\max(M_{\theta v})} = 0.6084$$

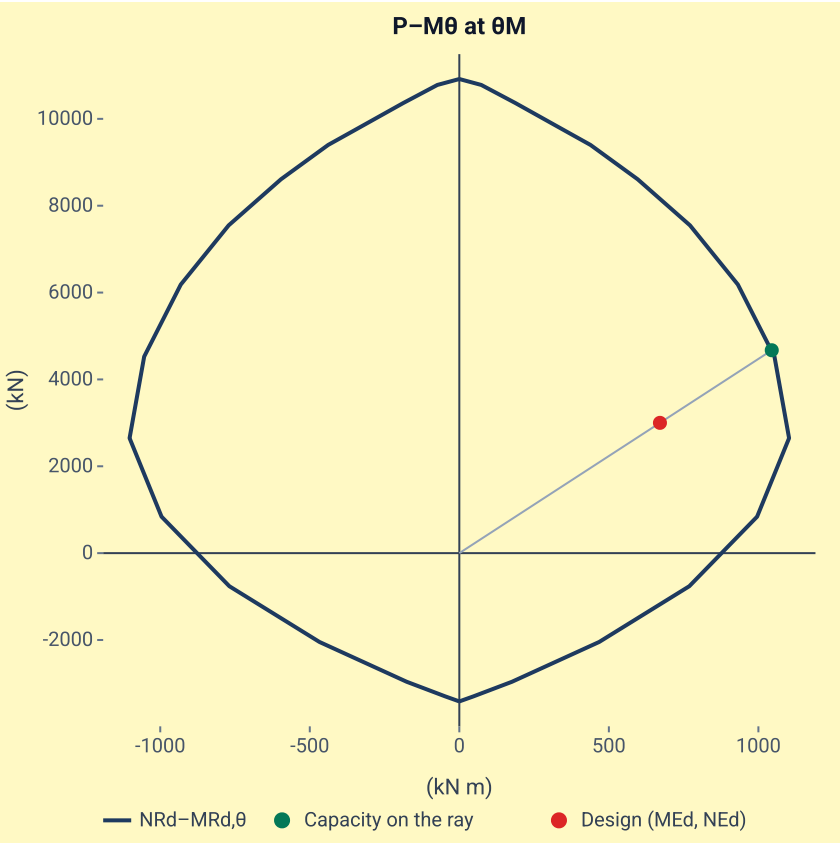
Distance to the curve along the ray (interpolated by direction)

$$\rho_{Rd} := \text{linterp}\left(\frac{n_v}{\sqrt{n_v^2+m_v^2}}, \sqrt{n_v^2+m_v^2}, \frac{n_{Ed}}{\sqrt{n_{Ed}^2+m_{Ed}^2}}\right) = 1.0396$$

Radial utilisation (governs) $$\text{UR}_M := \frac{\sqrt{n_{Ed}^2+m_{Ed}^2}}{\rho_{Rd}} = 0.6421$$

Check $$\text{UR}_M = 0.6421 \le 1$$ PASS

5.2 P–Mθ diagram



6. SHEAR CHECK (EN 1992-1-1:2023 §8.2.3, LINKS)

Truss in stresses: τEd = VEd/(bw·z) (8.18), links τRd,sy = ρw·fywd·cotθ (8.42), strut

σcd = τEd·(cotθ + tanθ) ≤ ν·fcd (8.44), ν = 0.5. cotθ: the largest value ≤ cot θmin (8.41)

that keeps σcd ≤ ν·fcd; cotθ < 1 (8.2.3(11)) never adds resistance and is not used.

Strength reduction for cracked concrete, 8.2.3(6) $\nu := 0.5$

Factor on cot θmin: 0.8 for class A links, 8.2.3(4) $k_A := 1 - 0.2\delta_A = 1$

Average axial stress, compression positive $\sigma_{cp} := \frac{N_{Ed}}{A_g} = 7.5 \cdot \mathrm{MPa}$

Area of one link leg $A_{sw1} := \frac{\pi \phi_w^2}{4} = 79 \cdot \mathrm{mm}^2$

Neutral-axis depth ratios for the compression chord search

$u_x := \mathrm{stack}\left(-0.1, \mathrm{seq}\left(0.05, 1, 0.05\right), 100\right)$

6.1 Shear along y (web b, depth h; bending about x)

Effective depth $d_y := h - c_{nom} - \phi_w - \frac{\phi}{2} = 738 \cdot \mathrm{mm}$

Lever arm, 8.2.1(3) $z_y := 0.9 d_y = 664 \cdot \mathrm{mm}$

Web width $b_{w.y} := b = 500 \cdot \mathrm{mm}$

Average shear stress (8.18)

$$\tau_{Ed.y} := \frac{V_{Ed.y}}{b_{w.y} z_y} = \frac{400 \cdot \mathrm{kN}}{(0.5 \cdot \mathrm{m}) \cdot (0.664 \cdot \mathrm{m})} = 1.21 \cdot \mathrm{MPa}$$

Compression chord depth x: neutral axis at the bending resistance under NEd (§8.1)

$$x_y := \mathrm{linterp}\left(\mathrm{map}\left(N_u, u_x, \frac{\pi}{2}\ \mathrm{rad}\right), h u_x, N_{Ed}\right) = 390 \cdot \mathrm{mm}$$

Flattest strut allowed (8.41)

$$\cot_{\theta min.y} := \begin{cases} \text{if } N_{Ed} < 0 \cdot \text{kN} \\ \quad \left\| \max\left(1, k_A\left(2.5 - 0.1 \cdot \frac{|N_{Ed}|}{V_{Ed.y}}\right)\right) \right. \\ \text{else if } x_y \geq 0.25 d_y \\ \quad \| 2.5 k_A \\ \text{else if } \sigma_{cp} \geq 3 \cdot \text{MPa} \\ \quad \| 3.0 k_A \\ \text{else} \\ \quad \left\| \left(2.5 + 0.5 \cdot \frac{\sigma_{cp}}{3 \cdot \text{MPa}}\right) k_A \right. \end{cases} = 2.5$$

Design cotθ: cot θmin if the strut holds, else the root of τEd(cotθ + tanθ) = ν·fcd; 1: strut fails

$$\cot_{\theta.y} := \begin{cases} \text{if } \tau_{Ed.y}\left(\cot_{\theta min.y} + \frac{1}{\cot_{\theta min.y}}\right) \leq \nu f_{cd} \\ \quad \| \cot_{\theta min.y} \\ \text{else if } 2\tau_{Ed.y} \leq \nu f_{cd} \\ \quad \left\| \frac{\nu f_{cd}}{2\tau_{Ed.y}} + \sqrt{\left(\frac{\nu f_{cd}}{2\tau_{Ed.y}}\right)^2 - 1} \right. \\ \text{else} \\ \quad \| 1 \end{cases} = 2.5$$

Area of links within s

$$A_{sw.y} := n_{w.y} A_{sw1} = 157 \cdot \text{mm}^2$$

Shear reinforcement ratio (8.43)

$$\rho_{w.y} := \frac{A_{sw.y}}{b_{w.y} s} = 0.0020944$$

Link resistance (8.42)

$$\begin{aligned} \tau_{Rd.sy.y} &:= \rho_{w.y} f_{ywd} \cot_{\theta.y} \\ &= 0.00209 \cdot (434.78 \cdot \text{MPa}) \cdot 2.5 \\ &= 2.28 \cdot \text{MPa} \end{aligned}$$

Strut stress (8.44)

$$\begin{aligned} \sigma_{cd.y} &:= \tau_{Ed.y}\left(\cot_{\theta.y} + \frac{1}{\cot_{\theta.y}}\right) \\ &= (1.21 \cdot \text{MPa}) \cdot \left(2.5 + \frac{1}{2.5}\right) \\ &= 3.5 \cdot \text{MPa} \end{aligned}$$

Shear utilisation ratio $\mathrm{UR}_{V.y} := \max\left(\frac{\tau_{Ed.y}}{\tau_{Rd.sy.y}}, \frac{\sigma_{cd.y}}{\nu f_{cd}}\right) = 0.5294$

Check $\mathrm{UR}_{\mathrm{V.y}} = 0.5294 \leq 1$ PASS

6.2 Shear along x (web h, depth b; bending about y)

Effective depth $d_x := b - c_{nom} - \phi_w - \frac{\phi}{2} = 438 \cdot \mathrm{mm}$

Lever arm, 8.2.1(3) $z_x := 0.9 d_x = 394 \cdot \mathrm{mm}$

Web width $b_{w.x} := h = 800 \cdot \mathrm{mm}$

Average shear stress (8.18)

$$\tau_{Ed.x} := \frac{V_{Ed.x}}{b_{w.x} z_x} = \frac{200 \cdot \mathrm{kN}}{(0.8 \cdot \mathrm{m}) \cdot (0.394 \cdot \mathrm{m})} = 0.635 \cdot \mathrm{MPa}$$

Compression chord depth x: neutral axis at the bending resistance under NEd (§8.1)

$x_x := \mathrm{linterp}\left(\mathrm{map}\left(N_u, u_x, 0 \cdot \mathrm{rad}\right), bu_x, N_{Ed}\right) = 243 \cdot \mathrm{mm}$

Flattest strut allowed (8.41)

$$\cot_{\theta min.x} := \begin{cases} \text{if } N_{Ed} < 0 \cdot \mathrm{kN} \\ \quad \max\left(1, k_A\left(2.5 - 0.1 \cdot \frac{|N_{Ed}|}{V_{Ed.x}}\right)\right) \\ \text{else if } x_x \geq 0.25 d_x \\ \quad 2.5 k_A \\ \text{else if } \sigma_{cp} \geq 3 \cdot \mathrm{MPa} \\ \quad 3.0 k_A \\ \text{else} \\ \quad \left(2.5 + 0.5 \cdot \frac{\sigma_{cp}}{3 \cdot \mathrm{MPa}}\right) k_A \end{cases} = 2.5$$

Design cotθ: cot θmin if the strut holds, else the root of τEd(cotθ + tanθ) = ν·fcd; 1: strut fails

$$\cot_{\theta.x} := \begin{cases} \text{if } \tau_{Ed.x}\left(\cot_{\theta min.x} + \frac{1}{\cot_{\theta min.x}}\right) \le \nu f_{cd} \quad = 2.5 \\ \quad \|\cot_{\theta min.x} \\ \text{else if } 2\tau_{Ed.x} \le \nu f_{cd} \\ \quad \left\| \frac{\nu f_{cd}}{2\tau_{Ed.x}} + \sqrt{\left(\frac{\nu f_{cd}}{2\tau_{Ed.x}}\right)^2 - 1} \right. \\ \text{else} \\ \quad \|1 \end{cases}$$

Area of links within s

$$A_{sw.x} := n_{w.x} A_{sw1} = 314 \cdot \text{mm}^2$$

Shear reinforcement ratio (8.43)

$$\rho_{w.x} := \frac{A_{sw.x}}{b_{w.x} s} = 0.00261799$$

Link resistance (8.42)

$$\begin{aligned} \tau_{Rd.sy.x} &:= \rho_{w.x} f_{ywd} \cot_{\theta.x} \\ &= 0.00262 \cdot (434.78 \cdot \text{MPa}) \cdot 2.5 \\ &= 2.85 \cdot \text{MPa} \end{aligned}$$

Strut stress (8.44)

$$\begin{aligned} \sigma_{cd.x} &:= \tau_{Ed.x}\left(\cot_{\theta.x} + \frac{1}{\cot_{\theta.x}}\right) \\ &= (634.92 \cdot \text{kPa}) \cdot \left(2.5 + \frac{1}{2.5}\right) \\ &= 1.84 \cdot \text{MPa} \end{aligned}$$

Shear utilisation ratio

$$\text{UR}_{V.x} := \max\left(\frac{\tau_{Ed.x}}{\tau_{Rd.sy.x}}, \frac{\sigma_{cd.x}}{\nu f_{cd}}\right) = 0.2231$$

Check

$$\text{UR}_{\text{V.x}} = 0.2231 \le 1$$ ✓ PASS

7. SUMMARY

Radial utilisation on P–Mθ (governs) $\text{UR}_M = 0.6421$

Shear along y $\text{UR}_{V.y} = 0.5294$

Shear along x $\text{UR}_{V.x} = 0.2231$

ANNEX

A.1 Bar layout and coordinates

Half-distance between corner bar centres along x

$$a_x := \frac{b}{2} - c_{nom} - \phi_w - \frac{\phi}{2} = 187 \cdot \text{mm}$$

Half-distance between corner bar centres along y

$$a_y := \frac{h}{2} - c_{nom} - \phi_w - \frac{\phi}{2} = 338 \cdot \text{mm}$$

Bar spacing (centres) along b

$$s_x := \frac{2a_x}{n_b - 1} = 125 \cdot \text{mm}$$

Bar spacing (centres) along h

$$s_y := \frac{2a_y}{n_h - 1} = 135 \cdot \text{mm}$$

Total number of bars

$$n_{bar} := 2n_b + 2\left(n_h - 2\right) = 16$$

Bars are numbered: top row (left to right), bottom row, left side, right side (corners once).

Bar index

$$k_s := \text{seq}\left(0, n_{bar} - 1\right)$$

x of each bar

$$x_s := \begin{cases} \text{if } k_s < n_b \\ \quad -a_x + k_s s_x \\ \text{else if } k_s < 2n_b \\ \quad -a_x + \left(k_s - n_b\right) s_x \\ \text{else if } k_s < 2n_b + n_h - 2 \\ \quad -a_x \\ \text{else} \\ \quad a_x \end{cases}$$

$$y_s := \left\| \begin{array}{l} \text{if } k_s < n_b \\ \quad \| a_y \\ \text{else if } k_s < 2n_b \\ \quad \| -a_y \\ \text{else if } k_s < 2n_b + n_h - 2 \\ \quad \| -a_y + (k_s - 2n_b + 1)\, s_y \\ \text{else} \\ \quad \| -a_y + (k_s - 2n_b - n_h + 3)\, s_y \end{array} \right.$$

y of each bar

Bar table

$$k_s = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \\ 13 \\ 14 \\ 15 \end{bmatrix} \quad x_s = \begin{bmatrix} -187 \\ -63 \\ 62 \\ 187 \\ -187 \\ -63 \\ 62 \\ 187 \\ -187 \\ -187 \\ -187 \\ -187 \\ 187 \\ 187 \\ 187 \\ 187 \end{bmatrix} \text{mm} \quad y_s = \begin{bmatrix} 338 \\ 338 \\ 338 \\ 338 \\ -338 \\ -338 \\ -338 \\ -338 \\ -203 \\ -68 \\ 68 \\ 203 \\ -203 \\ -68 \\ 68 \\ 203 \end{bmatrix} \text{mm}$$

A.2 Tie leg arrangement (EC2 9.5.3(6))

No inner ties when every bar is within 150 mm of a corner bar. Otherwise corners and every other bar are restrained, symmetric about the face centre.

Each mirrored pair is held by one inner closed tie; a single middle bar gets a cross-tie.

Every unrestrained bar is next to a restrained one, so a bar spacing ≤ 150 mm suffices.

$$l_b := \text{floor}\left(\frac{n_b - 1}{2}\right) s_x = 125 \cdot \text{mm}$$

Farthest bar from a corner on a b face

$$l_h := \text{floor}\left(\frac{n_h - 1}{2}\right) s_y = 270 \cdot \text{mm}$$

Farthest bar from a corner on an h face

Inner ties across the depth (pairs 2i and nb − 1 − 2i on the b faces)

$$q_b := \begin{cases} \text{if } l_b > 150 \cdot \text{mm} \\ \quad \left\| \text{floor}\left(\frac{n_b - 1}{4}\right) \right. \\ \text{else} \\ \quad \| 0 \end{cases} = 0$$

Inner ties across the width (pairs on the h faces)

$$q_h := \begin{cases} \text{if } l_h > 150 \cdot \text{mm} \\ \quad \left\| \text{floor}\left(\frac{n_h - 1}{4}\right) \right. \\ \text{else} \\ \quad \| 0 \end{cases} = 1$$

Last pair is a single middle bar (cross-tie, one leg): 1, else 0

$$m_b := \begin{cases} \text{if } q_b \geq 1 \wedge 4q_b = n_b - 1 \\ \quad \| 1 \\ \text{else} \\ \quad \| 0 \end{cases} = 0$$

$$m_h := \begin{cases} \text{if } q_h \geq 1 \wedge 4q_h = n_h - 1 \\ \quad \| 1 \\ \text{else} \\ \quad \| 0 \end{cases} = 0$$

Full-depth tie legs parallel to y (resist VEd,y) $n_{w.y} := 2 + 2q_b - m_b = 2$

Full-width tie legs parallel to x (resist VEd,x) $n_{w.x} := 2 + 2q_h - m_h = 4$

A.3 Concrete fibres

The concrete is cut into an nf × nf grid of fibres; each carries the stress at its centre strain.

Fibre index (row by row) $f := \text{seq}\left(0, n_f^2 - 1\right)$

Fibre centres $x_f := \left(f - n_f \text{floor}\left(\frac{f}{n_f}\right) + 0.5\right) \frac{b}{n_f} - \frac{b}{2}$

$$y_f := \left(\text{floor}\left(\frac{f}{n_f}\right) + 0.5\right)\frac{h}{n_f} - \frac{h}{2}$$

Area of one fibre $A_f := \frac{bh}{n_f^2} = 444 \cdot \text{mm}^2$

A.4 Tie drawing (data of the section figure)

Ties wrap the bars: each corner is an arc centred on the bar. Closed ties end in two 135° hooks at the top-left bar; cross-ties (legs) have a 135° hook at both ends.

Tie centreline radius round a bar $\delta_w := \frac{\phi + \phi_w}{2} = 18 \cdot \text{mm}$

Hook extension past the bend, EC2 Fig. 8.5 $L_w := \max(5\phi_w, 50 \cdot \text{mm}) = 50 \cdot \text{mm}$

Points drawn per arc $n_a := 7$

Inner closed ties across the depth (a middle bar is a leg) $Q_b := q_b - m_b = 0$

Inner closed ties across the width $Q_h := q_h - m_h = 1$

Closed ties: the outer tie and the inner ties $n_T := 1 + Q_b + Q_h = 2$

Points per closed tie: 5 arcs and 2 hook ends $M_T := 5n_a + 2 = 37$

Tie t (0 = outer): half-distance between its corner bars along x

$$b_T(t) := \begin{cases} \text{if } t \le Q_b \\ \quad \| a_x - 2t \cdot s_x \\ \text{else} \\ \quad \| a_x \end{cases}$$

Same along y

$$h_T(t) := \begin{cases} \text{if } t \le Q_b \\ \quad \| a_y \\ \text{else} \\ \quad \| a_y - 2(t - Q_b)\, s_y \end{cases}$$

Point j: its tie and its index within the tie $i_T(j) := \text{floor}\left(\frac{j}{M_T}\right)$

$$u_T(j) \coloneqq j - M_T i_T(j)$$

Arc of point j: 0 top-left hook, 1 TR, 2 BR, 3 BL, 4 top-left hook

$$a_T(j) \coloneqq \min\left(\max\left(\operatorname{floor}\left(\frac{u_T(j)-1}{n_a}\right),0\right),4\right)$$

Fraction along the arc

$$w_T(j) \coloneqq \frac{\min(\max(u_T(j)-1-n_a a_T(j),0),n_a-1)}{n_a-1}$$

Angle round the bar in degrees (clockwise)

$$\alpha_T(j) \coloneqq \left\| \begin{array}{l} \text{if } a_T(j)=0 \\ \quad \| 225-135 w_T(j) \\ \text{else if } a_T(j)=4 \\ \quad \| -180-135 w_T(j) \\ \text{else} \\ \quad \| 180-90 a_T(j)-90 w_T(j) \end{array} \right.$$

1 at the two hook ends

$$o_T(j) \coloneqq \left\| \begin{array}{l} \text{if } u_T(j)=0 \vee u_T(j)=M_T-1 \\ \quad \| 1 \\ \text{else} \\ \quad \| 0 \end{array} \right.$$

Centre of the bar the point wraps

$$X_c(j) \coloneqq \left\| \begin{array}{l} \text{if } a_T(j)=1 \vee a_T(j)=2 \\ \quad \| b_T(i_T(j)) \\ \text{else} \\ \quad \| -b_T(i_T(j)) \end{array} \right.$$

$$Y_c(j) \coloneqq \left\| \begin{array}{l} \text{if } a_T(j)=2 \vee a_T(j)=3 \\ \quad \| -h_T(i_T(j)) \\ \text{else} \\ \quad \| h_T(i_T(j)) \end{array} \right.$$

Tie points; hook ends point at 315° into the core

$$X_T(j) \coloneqq X_c(j) + \delta_w \cos\left(\frac{\pi \alpha_T(j)}{180}\right) + \frac{\sqrt{2}}{2} L_w o_T(j)$$

$$Y_T(j) \coloneqq Y_c(j) + \delta_w \sin\left(\frac{\pi \alpha_T(j)}{180}\right) - \frac{\sqrt{2}}{2} L_w o_T(j)$$

Points of all closed ties $j_T \coloneqq \text{seq}(0, n_T M_T - 1)$

Legs (cross-ties to a single middle bar) $n_L \coloneqq m_b + m_h = 0$

Points per leg: 2 hook arcs and 2 hook ends $M_L \coloneqq 2n_a + 2 = 16$

Point j: its leg and its index within the leg $i_L(j) \coloneqq \text{floor}\left(\frac{j}{M_L}\right)$

$$u_L(j) \coloneqq j - M_L i_L(j)$$

Leg direction: 0 parallel to y at x = 0, 1 parallel to x at y = 0 $k_L(j) \coloneqq i_L(j) + 1 - m_b$

Arc of point j (0 first bar, 1 second bar), fraction along it

$$a_L(j) \coloneqq \min\left(\max\left(\text{floor}\left(\frac{u_L(j) - 1}{n_a}\right), 0\right), 1\right)$$

$$w_L(j) \coloneqq \frac{\min(\max(u_L(j) - 1 - n_a a_L(j), 0), n_a - 1)}{n_a - 1}$$

Angle round the bar in degrees (two 135° arcs) $\alpha_L(j) \coloneqq 90 k_L(j) + 135 - 135 a_L(j) - 135 w_L(j)$

1 at the first / last hook end

$$o_s(j) \coloneqq \left\| \begin{array}{l} \text{if } u_L(j) = 0 \\ \quad \| 1 \\ \text{else} \\ \quad \| 0 \end{array} \right.$$

$$o_e(j) \coloneqq \left\| \begin{array}{l} \text{if } u_L(j) = M_L - 1 \\ \quad \| 1 \\ \text{else} \\ \quad \| 0 \end{array} \right.$$

Hook end directions (into the core) $$p_x(j) := (2k_L(j) - 1)\, o_s(j) - o_e(j)$$

$$p_y(j) := (1 - 2k_L(j))\, o_e(j) - o_s(j)$$

Leg points

$$X_L(j) := k_L(j)\,(2a_L(j) - 1)\, a_x + \delta_w \cos\left(\frac{\pi \alpha_L(j)}{180}\right) + \frac{\sqrt{2}}{2} L_w p_x(j)$$

$$Y_L(j) := (1 - k_L(j))\,(1 - 2a_L(j))\, a_y + \delta_w \sin\left(\frac{\pi \alpha_L(j)}{180}\right) + \frac{\sqrt{2}}{2} L_w p_y(j)$$

Points of all legs $$j_L := \mathrm{seq}(0, \max(n_L, 1) M_L - 1)$$

A.5 Strain plane, stress laws and section resultants

c = neutral-axis depth below the compression face; p = fibre coordinate from the centroid, positive towards the compression face; H = section depth in that direction; dt = depth of the extreme bar. The strain plane pivots about A (bar at −εud), B (face at εcu2) or C (εc2 at (1 − εc2/εcu2)·H), whichever governs for the depth c (EC2 Fig. 6.1). Compression positive.

Strain at fibre p

$$\varepsilon(c, p, H, d_t) := \begin{cases} \text{if } c < \frac{\varepsilon_{cu2} d_t}{\varepsilon_{cu2} + \varepsilon_{ud}} \\ \quad \varepsilon_{ud} \dfrac{c - \frac{H}{2} + p}{d_t - c} \\ \text{else if } c \le H \\ \quad \varepsilon_{cu2} \dfrac{c - \frac{H}{2} + p}{c} \\ \text{else} \\ \quad \varepsilon_{c2} \dfrac{c - \frac{H}{2} + p}{c - \left(1 - \frac{\varepsilon_{c2}}{\varepsilon_{cu2}}\right) H} \end{cases}$$

Concrete stress, parabola-rectangle (EC2 3.17), no tension

$$\sigma_c(\varepsilon) := \begin{cases} \text{if } \varepsilon \le 0 \\ \quad 0 \cdot \text{MPa} \\ \text{else if } \varepsilon < \varepsilon_{c2} \\ \quad f_{cd}\left(1 - \left(1 - \frac{\varepsilon}{\varepsilon_{c2}}\right)^2\right) \\ \text{else} \\ \quad f_{cd} \end{cases}$$

Steel stress, elastic-plastic within ±fyd

$$\sigma_s(\varepsilon) := \begin{cases} \text{if } E_s\varepsilon < -f_{yd} \\ \quad -f_{yd} \\ \text{else if } E_s\varepsilon > f_{yd} \\ \quad f_{yd} \\ \text{else} \\ \quad E_s\varepsilon \end{cases}$$

For a neutral-axis direction β (normal of the axis, towards the compression face):

Section depth along β

$$H_\beta(\beta) := b\,|\cos(\beta)| + h\,|\sin(\beta)|$$

Depth of the extreme bar

$$d_\beta(\beta) := \frac{H_\beta(\beta)}{2} - \min(x_s\cos(\beta) + y_s\sin(\beta))$$

Stress in every concrete fibre at depth c

$$\sigma_f(c,\beta) := \sigma_c(\varepsilon(c, x_f\cos(\beta) + y_f\sin(\beta), H_\beta(\beta), d_\beta(\beta)))$$

Strain in every bar

$$\varepsilon_b(c,\beta) := \varepsilon(c, x_s\cos(\beta) + y_s\sin(\beta), H_\beta(\beta), d_\beta(\beta))$$

Net bar stress (concrete it displaces subtracted)

$$\sigma_b(c,\beta) := \sigma_s(\varepsilon_b(c,\beta)) - \sigma_c(\varepsilon_b(c,\beta))$$

Axial resistance

$$N_R(c,\beta) := A_f\text{sum}(\sigma_f(c,\beta)) + A_b\text{sum}(\sigma_b(c,\beta))$$

Moment about x (lever arm y)

$$M_{Rx}(c,\beta) := A_f\text{sum}(\sigma_f(c,\beta)\,y_f) + A_b\text{sum}(\sigma_b(c,\beta)\,y_s)$$

Moment about y (lever arm x)

$$M_{Ry}(c,\beta) := A_f\text{sum}(\sigma_f(c,\beta)\,x_f) + A_b\text{sum}(\sigma_b(c,\beta)\,x_s)$$

A.6 P–Mθ diagram data (neutral-axis sweep)

$$\beta_v := \left(\frac{\pi}{2}\,\text{floor}\left(\frac{2\theta_M}{\pi\,\text{rad}}\right) + \frac{\pi}{18}\,\text{seq}(0,9)\right)\,\text{rad}$$

Angles β swept over the quadrant of θ (10° steps)

P–Mθ curve

For each depth c = u·H(β) the moment direction ψ is found for every β of the sweep; the point with ψ = θM (interpolated between the β steps) is the point of the P–Mθ curve.

At c = u·H(β): N, M and direction ψ $N_u(u,\beta) := N_R(uH_\beta(\beta),\beta)$

$$M_u(u,\beta) := \sqrt{(M_{Rx}(uH_\beta(\beta),\beta))^2 + (M_{Ry}(uH_\beta(\beta),\beta))^2}$$

Moment direction, measured from β (no 0/360° jump)

$$\psi_u(u,\beta) := \beta + \arctan\left(\frac{M_{Rx}(uH_\beta(\beta),\beta)\cos(\beta) - M_{Ry}(uH_\beta(\beta),\beta)\sin(\beta)}{M_{Ry}(uH_\beta(\beta),\beta)\cos(\beta) + M_{Rx}(uH_\beta(\beta),\beta)\sin(\beta)}\right)$$

Curve point for u (ψ = θM) $N_\theta(u) := \text{linterp}(\text{map}(\psi_u,u,\beta_v),\text{map}(N_u,u,\beta_v),\theta_M)$

$$M_\theta(u) := \text{linterp}(\text{map}(\psi_u,u,\beta_v),\text{map}(M_u,u,\beta_v),\theta_M)$$

$$\beta_\theta(u) := \text{linterp}(\text{map}(\psi_u,u,\beta_v),\beta_v,\theta_M)$$

Control points u = c/H (the ends, pure tension and uniform εc2, have M = 0)

$u_t := \text{stack}(0.02, 0.05, \text{seq}(0.1, 1.1, 0.1), 1.5, 3)$

P–Mθ curve

$N_{\theta v} := \text{stack}(N_R(-0.1H_\beta(\theta_M),\theta_M),\text{map}(N_\theta,u_t),N_R(100H_\beta(\theta_M),\theta_M))$

$M_{\theta v} := \text{stack}(0\cdot\text{kN}\cdot\text{m},\text{map}(M_\theta,u_t),0\cdot\text{kN}\cdot\text{m})$

u of every curve point $u_c := \text{stack}(-0.1, u_t, 100)$

P–Mθ control points

$$N_{\theta v} = \begin{bmatrix} -3415 \\ -3406 \\ -3299 \\ -2962 \\ -2045 \\ -762 \\ 840 \\ 2648 \\ 4526 \\ 6184 \\ 7547 \\ 8609 \\ 9399 \\ 9747 \\ 10361 \\ 10781 \\ 10917 \end{bmatrix} \text{kN} \qquad M_{\theta v} = \begin{bmatrix} 0 \\ 3 \\ 47 \\ 176 \\ 468 \\ 769 \\ 996 \\ 1103 \\ 1054 \\ 931 \\ 771 \\ 596 \\ 439 \\ 348 \\ 188 \\ 73 \\ 0 \end{bmatrix} \text{kN} \cdot \text{m}$$